Comparative Analysis of Advanced CNN Architectures for Multi-Class Brain Tumor Classification Using MRI

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***Abstract*—Early detection and planning for treatment of brain tumors is essential and is facilitated by magnetic resonance imaging (MRI).. Currently, manual reading of MRI is labor intensive and can be subject to mistakes, since various types of tumors share similar visual features. In this research, a deep learning system using AI technique for the classification of brain tumors into multiple classes using MRI images is presented. The data set in this study includes four classes: glioma, meningioma, pituitary and normal brain MRI scans. Tumor classification was performed using the advanced pretrained deep learning architectures implemented employing transfer learning and fine tuning techniques such as EfficientNetV2, DenseNet121, and ResNet50V2. The accuracy, precision, recall, F1-score, ROC-AUC score, confusion matrix, and classification report were used to evaluate the models. Experimental results demonstrated highly accurate classification, with EfficientNetV2 achieving an accuracy of 98.92% and the best Macro-AUC score (0.9999) and the least misclassification rate among all the tumor classes. Furthermore, Grad-CAM visualization was employed to interpret the model and enhance the interpretability of the model by visualizing regions of interest that focused on the tumor and led to predictions. The proposed system exhibits good robustness, reliability, and effectiveness in automated brain tumor diagnosis, and also shows the potential of deep learning techniques in the application of AI medical image processing.**

Keywords: Brain Tumor Detection, Deep Learning, MRI Images, EfficientNetV2, Transfer Learning, Grad-CAM

***Index Terms*—Brain Tumor Detection, Deep Learning, MRI Images, EfficientNetV2, Transfer Learning, Grad-CAM**

I. INTRODUCTION

Brain tumors are among the most serious neurological diseases and rapid and accurate diagnosis is necessary for a successful treatment and better patient outcome. Magnetic Resonance Imaging (MRI) is the most popular method for diagnosing brain tumors, as the images it produces reveal the details of the brain's tissues. However, manual assessment of MRI scans can be quite time-consuming and subject to human errors especially in the case of similar visual qualities of different types of tumors. With the progress of Artificial Intelligence (AI) and Deep Learning, automated brain tumor classification has emerged as an important area of research.

For the proposed study, the advanced pretrained deep learning architectures such as EfficientNetV2, DenseNet121 and ResNet50V2 are used for multiclass classification of brain tumor using MRI images. The MRI scans are divided into four classes: glioma, meningioma, pituitary tumor and normal brain. Transfer learning, fine-tuning, and data augmentation methods are used to enhance classification accuracy and generalization of the model. Furthermore, Grad-CAM visualization is used to localize the regions of the image that have the greatest influence on the model's decisions, helping to improve the interpretability of the model.

While various deep learning models have been proposed for brain tumor classification, only a few studies give a comprehensive comparison of the latest pretrained CNN architectures in the same experimental setting. Moreover, the majority of the existing research does not adequately analyze model interpretability and only considers classification accuracy. In order to fill these gaps, this study proposes a comparative analysis of EfficientNetV2, DenseNet121 and ResNet50V2 performing the evaluation on the same dataset, using the same preprocessing pipeline, the same transfer learning strategy and the same evaluation metrics. Furthermore, Grad-CAM is used to enhance the interpretability of the proposed models to make the process of computer-aided diagnosis more transparent and reliable.

II. LITERATURE REVIEW

[1] Two deep learning models and six conventional machine learning models were evaluated and their performance compared for automatic brain tumor detection and classification based on 3,264 MRI images of gliomas, meningiomas, and pituitary tumors as well as healthy brain images. Crop rotation and flipping were used on the data prior to training. The proposed 2D-CNN model achieved the excellent accuracy of 96.47% during training and the accuracy of 93.45% during testing with a good ROC-AUC performance (0.99–1). By contrast, the traditional machine learning models recorded lower performance with the highest accuracy of 86% in KNN. The study showed that deep learning models are much

more effective in brain tumor detection and classification than traditional machine learning techniques.

[2] The paper presents an automatic deep learning based system for brain tumor detection and classification using magnetic resonance imaging (MRI) image of brain. The MRI images were classified to four classes (glioma, meningioma, pituitary tumor, and normal brain) using a CNN model trained with a pre-trained VGG16 network. The study was based on 17,136 MRI images taken from various public datasets and data augmentation techniques. Effective fine-tuning, network optimization, and dataset diversity led to a high classification accuracy of 99.24% by the proposed model. The authors were also able to create a web-based upload system for MRI images and a prediction system for tumors to facilitate clinical diagnosis quickly and accurately.

[3] This review article elaborates on automatic brain tumor segmentation techniques with the help of MRI images in the field of early diagnosis and treatment planning. The existing methods are grouped into classical methods, machine learning techniques, and deep learning models to separate abnormal brain tissues from normal tissues. It also outlines some commonly used datasets, CNN architectures, and the autoencoder based models where most of the time these models provide better results as compared to traditional statistical and machine learning models in brain tumor segmentation. The paper also highlights the existing obstacles and future research avenues in the design of accurate, robust and automated brain tumour segmentation systems.

[4] The study proposes an automated system to classify brain tumor based on MRI images by applying four pre-trained CNN architectures: Xception, NasNet Large, DenseNet121 and InceptionResNetV2. Two public datasets are used to generate MR images and irrelevant regions are cropped off, then normalization is performed and images are subject to extensive augmentation (rotation, flipping, translation) to increase the amount of medical images and minimize the risk of overfitting. The features are extracted from the pre-trained networks and fed to the added dense layers, where they are trained using regularization, dropout, batch normalization and Adam optimizer for the classification of images as tumors or normal. Of the models, the Xception model trained using Adam achieved the highest accuracy of 99.67MRI dataset and 91.94other models tested and previous studies, indicating that the model has a great potential for rapid and accurate computer- aided diagnosis of brain tumors from MRI images.

[5] This paper examines the performance of various pre-trained deep learning models for automatic brain tumour classification using 7,023 MRI images, belonging to the classes of glioma, meningioma, pituitary tumour and normal brain. The models like Xception, MobileNetV2, InceptionV3, ResNet50, VGG16, and DenseNet121 were used, and transfer learning, preprocessing, and data augmentation techniques were used to improve the classification performance of these models. The highest overall accuracy was obtained by Xception, which reached a score of98.7% and a weighted F1 score of 95.3among all the models. The study also emphasized problems with decreased recall for glioma and meningioma tumors and the lack of explainability in deep learning models.

[6] This paper is a review of the use of explainable artificial intelligence (XAI) applied to deep learning (DL) medical image classification. Highly accurate "black-box" AI models are often hard for clinicians to trust, as clear explanations and transparency are necessary for medical decisions, the study highlights. The authors describe various methods of explanation (such as visual, textual, example-based, concept-based explanations) and interpretable and post-hoc AI models. The survey highlights that explainable AI is a crucial element in enhancing the reliability, transparency, and clinical acceptance of deep learning systems in medical imaging applications.

[7] The paper describes the applications of modern machine learning (ML) and deep learning approaches (DL) using MRI for better analysis of brain tumors. It stresses on the fact that brain tumor is extremely challenging to diagnose and elucidates why MRI is the preferred choice of imaging due to its high contrast and safety. It reviews several studies on many publicly available datasets like BraTS, TCIA, or Figshare, for various tasks like tumor detection, segmentation, classification and prediction of survival. It concludes that ML and DL methods can greatly aid brain tumor analysis, which enhances the accuracy of diagnosing brain tumors and developing personalized treatment plans.

[8] This research aims to compare the automatic brain tumor classification method based on deep learning and traditional machine learning techniques in the classification of brain tumors into categories of glioma, meningioma, pituitary tumor and healthy brain, using MRI images. The authors created a custom 2D-CNN and a convolutional autoencoder model, and benchmarked them with traditional models including KNN, SVM, Random Forest, Logistic Regression, SGD and MLP. The accuracy of the 2D-CNN obtained was around 96%/95% in training and recall respectively, which is significantly higher than the traditional machine learning methods where KNN obtained the highest training accuracy of 86%. The study has found that the CNN model proposed in the study is able to provide accurate and fast support for the detection of brain tumors in clinical practice.

[9] In this paper, we propose an automatic brain tumour classification system based on the combination of features extracted from the pretrained deep neural networks and classical machine learning classifiers. A total of 13 pretrained CNN models were used for the study as feature extractors of brain MRI images, such as ResNet, DenseNet, Inception, ResNeXt, MobileNet, and ShuffleNet. Multiple classifiers including SVM, KNN, Random Forest, AdaBoost and Gaussian Na¨ ıve Bayes were used to evaluate the extracted deep features. The results of the experiments confirmed that the features derived from the different CNN models gave significant improvement in classification performance and DenseNet derived features gave

the best performance, and SVM with RBF kernel worked as the most effective classifier which could classify brain tumor.

[10] This study is aimed towards the construction of a Hybrid Artificial Intelligence (AI) system to classify brain Magnetic Resonance Imaging (MRI) image as a meningioma tumour or healthy tissue. The authors compare the following: (1) original CNN architecture with last layer SoftMax; (2) hybrid architecture with the CNN feature extraction layer, followed by traditional machine learning classifiers including Random Forest, K-Nearest Neighbors and Support Vector Machine (SVM). The MRI scan is used with the EfficientNetB0-SVM hybrid model to attain the highest performance, where an accuracy of 99.5% is achieved in the classification of tumours from healthy brains. The study uses bagging ensemble method to increase the generalization ability and avoid overfitting.

[11] The aim of this paper is to propose an automated machine learning framework for brain tumour diagnosis based on MRI images to decrease the time-consuming manual analysis, which is performed by radiologists. The framework involves these preprocessing steps: Gaussian filtering, skull stripping, Histogram Equalization, and intensity normalization of the original MRI, and then segmentation by adaptive thresholding, edge detection, morphological processing and clustering of the image to identify the regions of the tumor. Random Forest (RF) and Support Vector Machine (SVM) models were used to classify the segmented MRI images into the glioma, meningioma, pituitary adenoma and no tumor categories. The study employed 5,712 MRI scans to train the algorithm and 1,311 MRI scans for testing, showing the algorithm's ability to achieve effective automated brain tumor classification performance.

[12] Within this research, influence of MRI radiomics preprocessing methods on the classification of PD in machine learning algorithms was examined. The study assessed various pre-processing pipelines, acquired radiomic features in various brain regions using T1-weighted MRI scans. The stability and repeatability of processing method were evaluated, and the results show that wavelet-based and texture features were the most repeatable and stable across different processing methods. High AUC scores were achieved in classifiers using best processing pipeline and also the removal of non-reproducible features allowed to enhance less efficient classifiers. It was shown that by carefully pre-processing and selecting the stable radiomic features, the robustness and accuracy of the subtype classification of PD could be enhanced.

[13] This paper introduces a two stage computerized method for better detection of brain tumors in a faster and accurate way by using MRI images. The first part is concerned about enhancing the quality of the MRI image, taking measures to reduce noise, enhance the contrast, and normalize it into better images. The second step involves classification via Support Vector Machine (SVM) of the obtained segmentation for each tumor category (meningioma, glioma and pituitary). When evaluated in a large, contrast-enhanced MRI database the accuracy of the system was demonstrated to be very high sensitivity, specificity, accuracy and Dice score; in addition, it achieves great processing speed compared to several existing methods for diagnosis of brain tumors.

[14] The aim of this study was to assess the effect of different MRI preprocessing pipelines on CNN-based brain tumor classification both in terms of accuracy and efficiency. Different pre-processing techniques, including skull stripping, contrast enhancement, bias field correction and denoising, were applied and compared in order to evaluate three different pre-processing methods: the standard method, the enhanced method, and the fast method. All pipelines gave comparable classification accuracy and F1-scores, significantly different compute time. This fast preprocessing pipeline needed fewer training durations, underlining the fact that a complex preprocessing does not always lead to significant gains and that quicker preprocessing procedures can be used to achieve a better compromise between benefits and training efficiency.

[15] The author gives survey on the uses of data augmentation in enhancing the deep learning capabilities in medical imaging stateful diagnostics for the cases of small or imbalanced image datasets. In medicine, the limited data is important due to the small number of patients, privacy problems and equipment limitations: this can lead to biased models and overfitting, which decreases the diagnostic accuracy. The study investigates various augmentation methods for brain, lung, breast and eye diseases with MR, CT, mammography and fundoscopy images, as well as how these popular augmentation methods affect performance of the deep learning classification. Performance results indicate that when appropriately customized for the specific image type and diagnosis of interest, data augmentation can make strides in improving performance.

[16] The paper provides a summary of 83 papers based on the CNN's used for brain tumor segmentation on MRI images published from 2015 to 2022. The review focuses on the datasets, the classification tasks assessed, the validated CNN architectures, as well as the performance measures adopted in these studies. The reported accuracy rates for classification of meningioma, glioma and pituitary tumors were high at about 92-100% and for differentiation of low-grade and high-grade gliomas the reported rates varied between 60-99%. The study showed that there is no single superior CNN architecture due to various datasets, validation and evaluation methods used in the studies. The authors also noted that only few studies investigated the clinical applicability and underscored future directions in translation of CNN-based brain tumor classification in real clinical practice.

III. METHODOLOGY

The proposed methodology is mainly based on the development of an AI based deep learning system for multiclass brain tumour classification using MRI image. The workflow covers dataset preparation, data augmentation, transfer learning with EfficientNetV2, model training and evaluation.

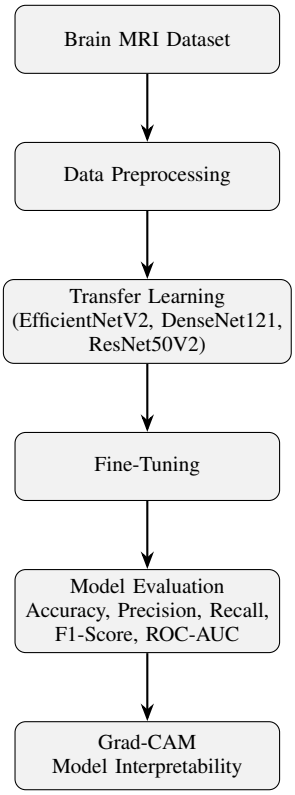


Fig. 1. Overall workflow of the proposed brain tumor classification framework.

A. Dataset Preparation

The brain Tumor MRI Dataset used was a public dataset from Kaggle [17]. Brain MRI images are represented in four classes: Glioma Tumor, Meningioma Tumor, Pituitary Tumor, and Normal Brain. The data set was divided into training and test sets, keeping the same class distribution in both sets, but not across all classes.

The split consisted of 80 percent training and 20 percent testing, and the model parameters were learned using the 80 percent of the MRI images.The training set was used to learn the model parameters, and the 20 percent of the MRI images was used to evaluate the model on unseen MRI images. All of the images were resized to 224×224 pixels prior to training to meet the input dimensions needed by the pretrained CNN architectures. Normalizing the pixel intensity values to the interval [0,1] helps improve the training stability and convergence.

Random horizontal flipping, random rotation, zooming, width shifting, height shifting and brightness adjustment were performed only on the training images to increase the generalization ability of the model and to prevent overfitting. The testing set was not augmented to provide a fair test of the proposed models.

B. Data Preprocessing

The data visualization step was conducted to study the diversity and characteristics of the MRI images used in this

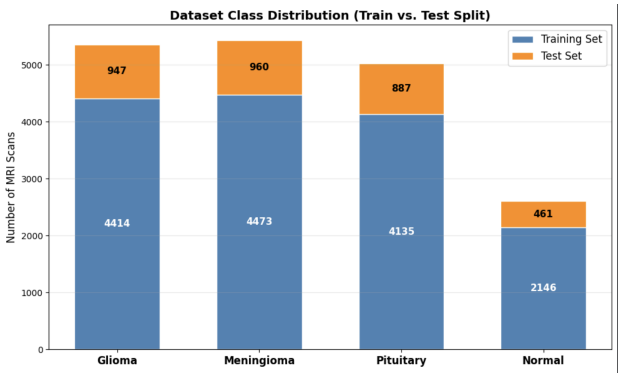
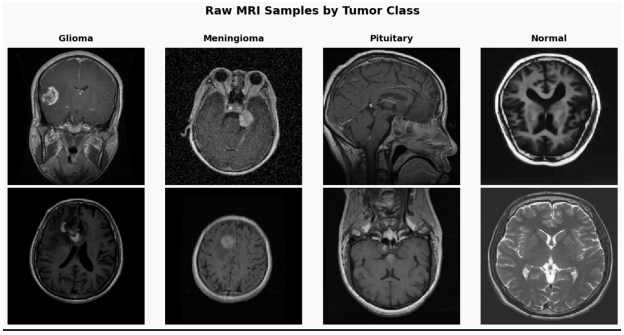


Fig. 2. Dataset Class distribution (Train vs. Test Spilt)

study. Some MRI scans of various tumor types, such as glioma, meningioma, pituitary and normal brain images were seen to gain insight into variations in the MRI dataset. The images displayed various differences in shape, size, orientation, brightness, and intensity levels of the tumors. The differences can illustrate the complexity of brain tumor classification, and show the need for preprocessing techniques before training the deep learning models.



Caption: Raw MRI Samples: Tumor Class

C. Proposed Model Architecture

The proposed system is based on a pretrained Convolutional Neural Network (CNN) model known as EfficientNetV2, which is a variant of transfer learning. EfficientNetV2 was chosen for its excellent capability to extract features, and its high performance in image classification tasks. The architecture is based on the EfficientNetV2 backbone, Global Average Pooling, Global Max Pooling, Batch Normalization, Dense layers, Dropout layers and a Softmax output layer with 4 classes for brain tumor classification. To enhance the efficiency of learning and classification, the model was initialized with the weights pre-trained on ImageNet.

D. Transfer Learning and Fine-Tuning

The training programme was split into two stages. During the first phase, the EfficientNetV2 backbone was not trained and only the newly added classification layers were trained. The final 50 layers of EfficientNetV2 were unfrozen during the second phase and optimized with a smaller learning

rate. This transfer learning and fine-tuning strategy enhances the ability to extract features and minimizes overfitting and overgeneralization of the model.

E. Optimization Techniques

The performance and stability of the models were optimized using several optimization techniques. The techniques used were: weight decay regularization, dropout layers, label smoothing, cosine decay learning rate scheduling, early stopping and class weighting for class imbalance. The methods will help to stabilize learning process, reduce overfitting and model's overall generalization.

F. Test-Time Augmentation (TTA)

Test-Time Augmentation (TTA) technique was used during the test to enhance the robustness of the prediction and accuracy of the classification. Several copies of each MRI were obtained and the final prediction was determined by taking the average prediction of all the augmented copies. This method enables to enhance the accuracy of the final class results.

G. Performance Evaluation

The model was trained and tested with various performance indicators such as accuracy, precision, recall, F1 score, ROC-AUC score, confusion matrix and classification report. The effectiveness and reliability of the proposed multiclass brain tumor classification system was analyzed using these evaluation metrics.

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \tag{1}$$

$$\text{Precision} = \frac{TP}{TP + FP} \tag{2}$$

$$\text{Recall} = \frac{TP}{TP + FN} \tag{3}$$

$$\text{F1 Score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \tag{4}$$

$$\text{TPR} = \frac{TP}{TP + FN} \tag{5}$$

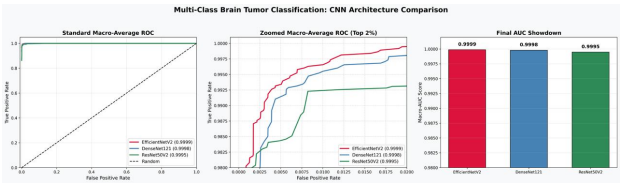
$$\text{FPR} = \frac{FP}{FP + TN} \tag{6}$$

$$AUC = \int_0^1 \text{TPR}(x)\, dx \tag{7}$$

$$\text{Macro-Precision} = \frac{1}{N} \sum_{i=1}^{N} \text{precision}_i \tag{8}$$

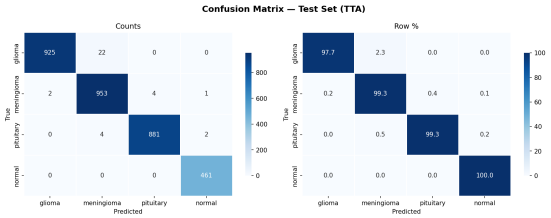
IV. EXPERIMENTATION AND RESULTS

The experimentation phase was dedicated to assess and compare the performance of various architectures of deep learning in comparison with the multiclass brain tumor classification problem using MRI images. To train and test the pre-trained deep learning models, three models were used, they are EfficientNetV2, DenseNet121 and ResNet50V2 and the prepared MRI dataset was used. For the evaluation of the classification capability and robustness of the models, ROC (Receiver Operating Characteristic) analysis and Macro-AUC scores were used.



The paper introduces a comparison of ROC and Macro-AUC of deep learning models. In this paper, comparative study with deep learning models based on ROC and Macro-AUC is presented.

To compare the performance of EfficientNetV2, DenseNet121 and ResNet50V2, we show the comparative performance analysis using Macro-AUC and macro-average ROC curve in Figure 3. Overall, the standard ROC curves are located near the upper-left corner, which suggests a very good classification performance and low false positive rates for glioma, meningioma, pituitary, and normal brain MRI images. The zoomed ROC analysis for the top 2% false positive region shows that EfficientNetV2 consistently outperforms DenseNet121 and ResNet50V2 in terms of true positive rate, highlighting its superior sensitivity and robustness under these strict classification conditions. Compared to DenseNet121 and ResNet50V2, both of which scored 0.9995, the score of EfficientNetV2 is the highest i.e. 0.9999 for the final bar graph representing the overall Macro-AUC scores. The results show that all three models achieved almost perfect classification accuracy, but the EfficientNetV2 model showed the highest overall accuracy and generalization ability.



The confusion matrix of the proposed efficientnetV2 model.

The classification performance for each tumor category is shown in the confusion matrix of the proposed EfficientNetV2 model. There were only a small number of misclassifications made, mainly with the images of gliomas and meningiomas, and high prediction accuracy was achieved in all

four MRI classes (glioma, meningioma, pituitary and normal). The classification accuracy of the normal brain class was 100%, demonstrating the superiority of the proposed model in discriminating normal brain.

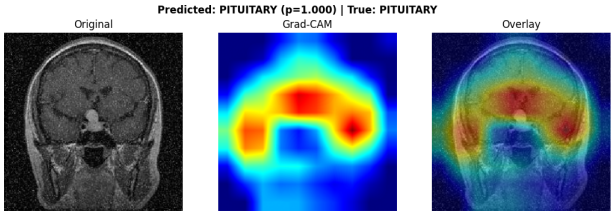


Fig. 3. Brain Tumor Classification using Grad-CAM Visualization

Using Grad-CAM visualization, the most important regions in MRI images that affected the model prediction were identified. The heatmaps revealed that the models primarily concentrated on the regions of the tumour when making classification decisions, enhancing the interpretability and transparency of the proposed AI system. The experimental results showed that all three deep learning models have good classification results: EfficientNetV2 has the highest overall accuracy in the classification of tumors.

V. CONCLUSION

This study introduced an artificial intelligence (AI) based deep-learning system to classify brain tumor into several classes, based on magnetic resonance imaging (MRI) images. The proposed system's model used three pre-trained deep learning architectures; EfficientNetV2, DenseNet121, ResNet50V2, which were trained to classify MRI images into 4 classes, normal brain, meningioma, pituitary tumor, and glioma. Data preprocessing, data augmentation, transfer learning and fine-tuning were used to enhance the generalization and the performance of the model. The experimental results showed good classification accuracy and excellent ROC-AUC results, with EfficientNetV2 achieving the highest score of Macro-AUC of 0.9999. The confusion matrix and evaluation metrics showed that there was little misclassification in the model, and Grad-CAM visualization improved the interpretability of the model by visualizing the important parts of the image for the model's prediction. As a whole, the proposed system was found to be feasible, reliable and effective in automatic classification of brain tumors from MRI images.

The proposed framework was successful in the sense of performance, but there are some limitations in this study. The experiments were performed with one public brain MRI dataset, which might not capture the full range of clinical data from various hospitals and imaging modalities. So the models need to be further validated using multi-institutional and larger datasets to assess their generalizability. Furthermore, future research can involve comparing the evaluated CNN architectures with recently developed Vision Transformer (ViT)-based and hybrid models of CNN–Transformer for exploring their performance in multiclass brain tumor classification.

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